

Adapting Kernel Security Testing to HPC Systems

Ryan Adamson

2026 HPC Testathon @ Cray User Group 2026

April 25, 2025 – Nice, France

github.com/rmadamson

* Supplemented with wisdom from the movie *Hackers*

ORNL is managed by UT-Battelle, LLC for the US Department of Energy



U.S. DEPARTMENT OF
ENERGY

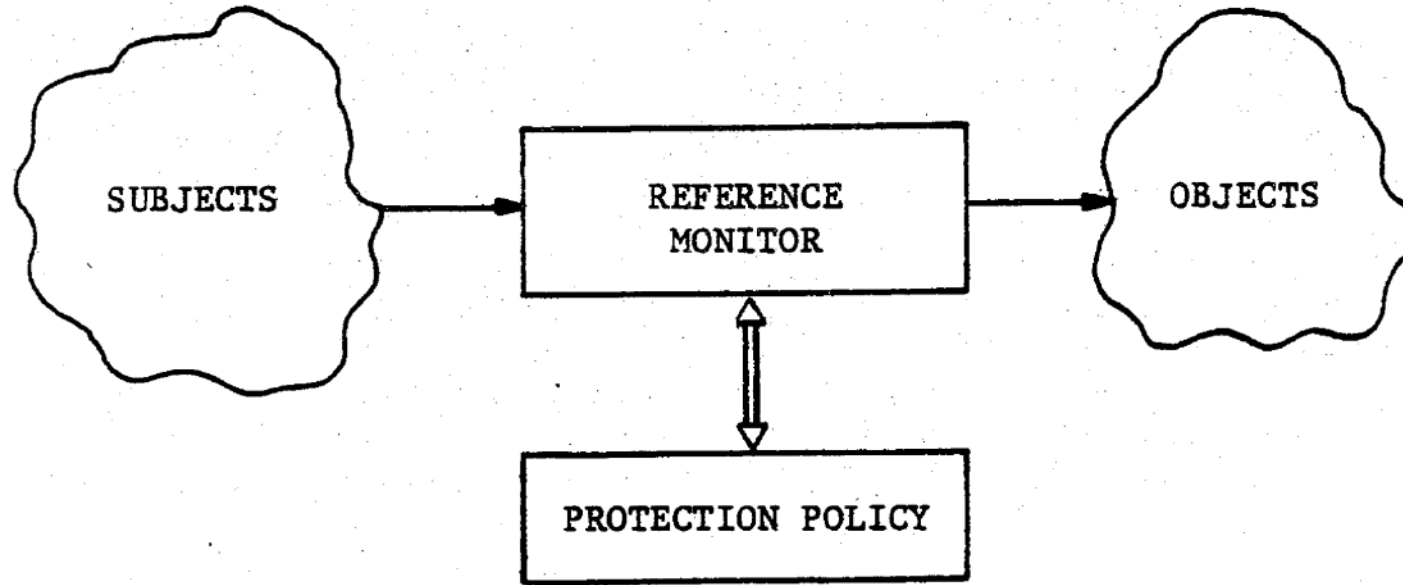


Figure 1. A Reference Monitor

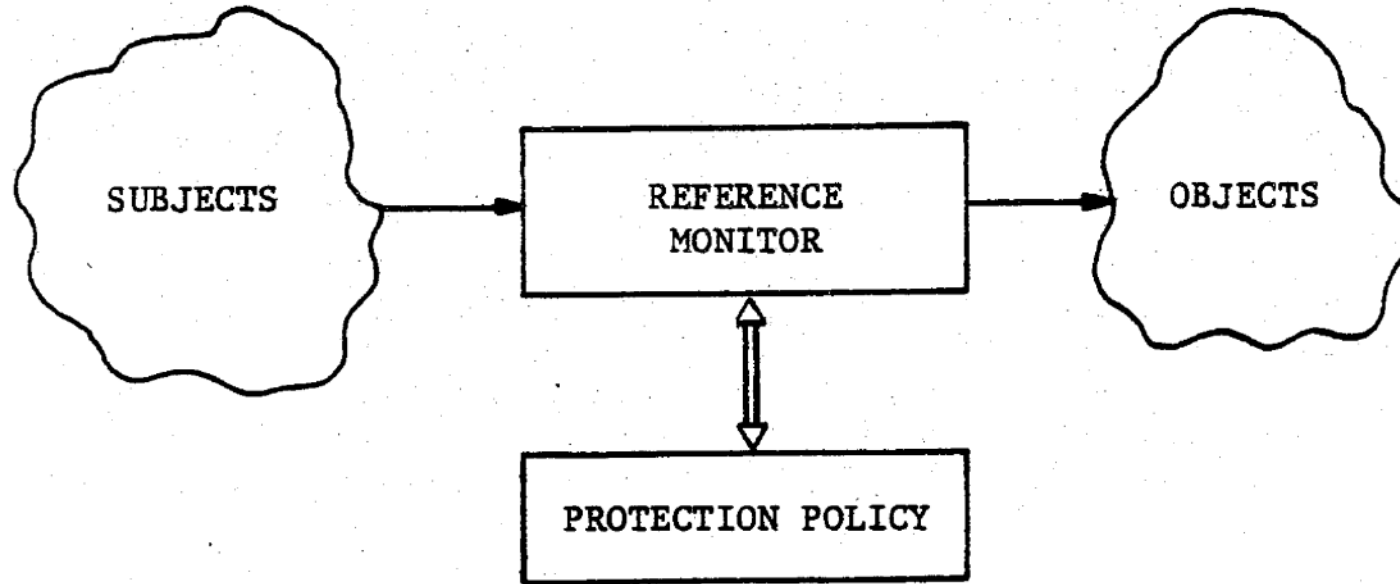


Figure 1. A Reference Monitor

Kenneth J. Biba: "Integrity Considerations for Secure Computer Systems", MTR-3153, The Mitre Corporation, April 1977

A reference monitor must satisfy three logical properties:

1) it is complete: all accesses by subjects to objects are monitored and enforced;

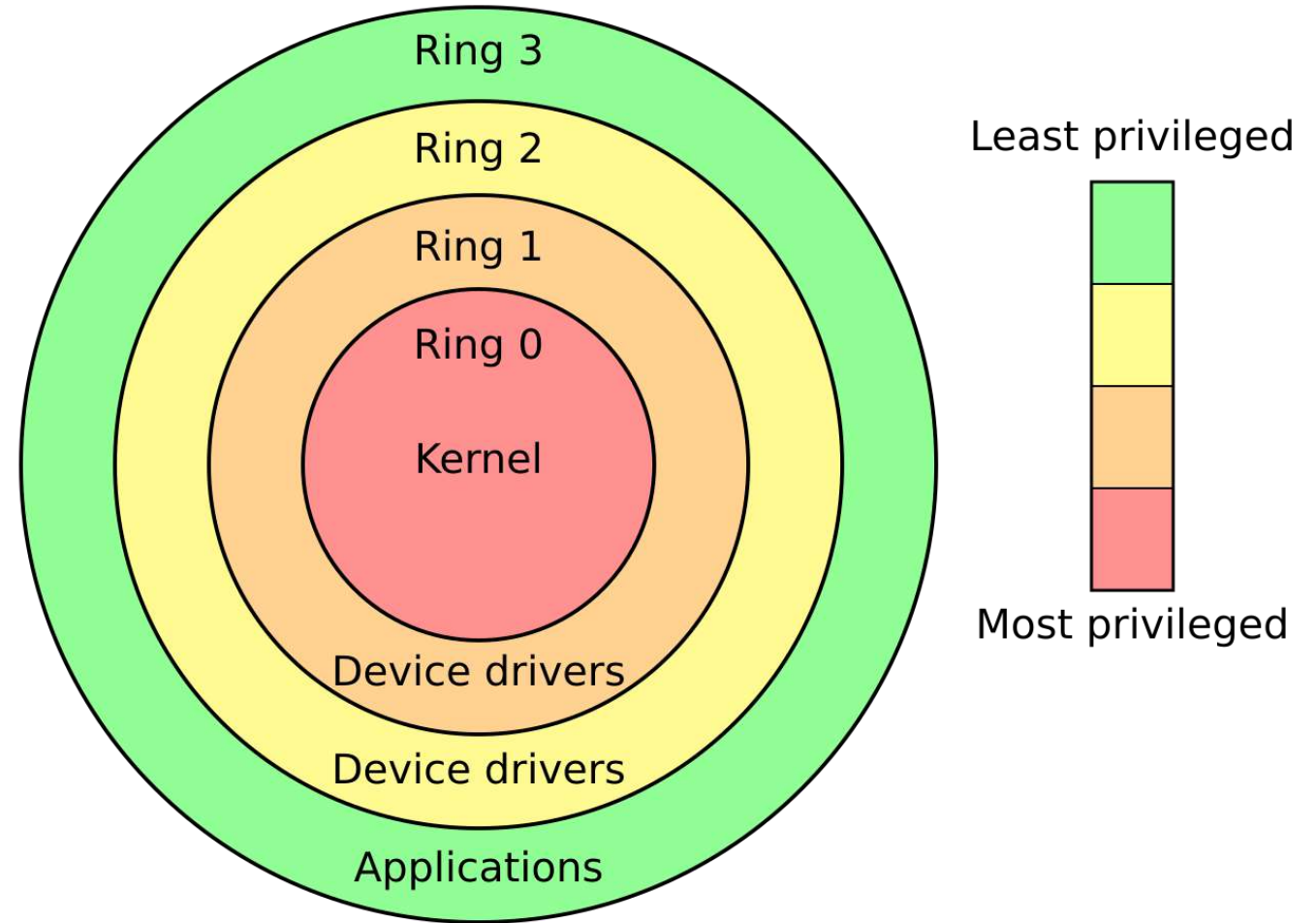
2) it is protected: its function may not be maliciously or accidentally modified by unauthorized forces; and

3) it has provably proper behavior: it must faithfully enforce the specified protection policy.

Linux Security Kernel Model (Reference Monitor)

• Protection Rings

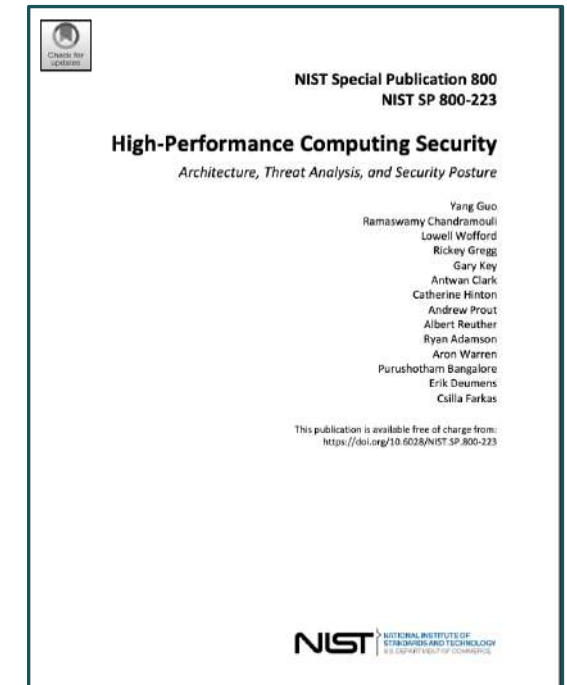
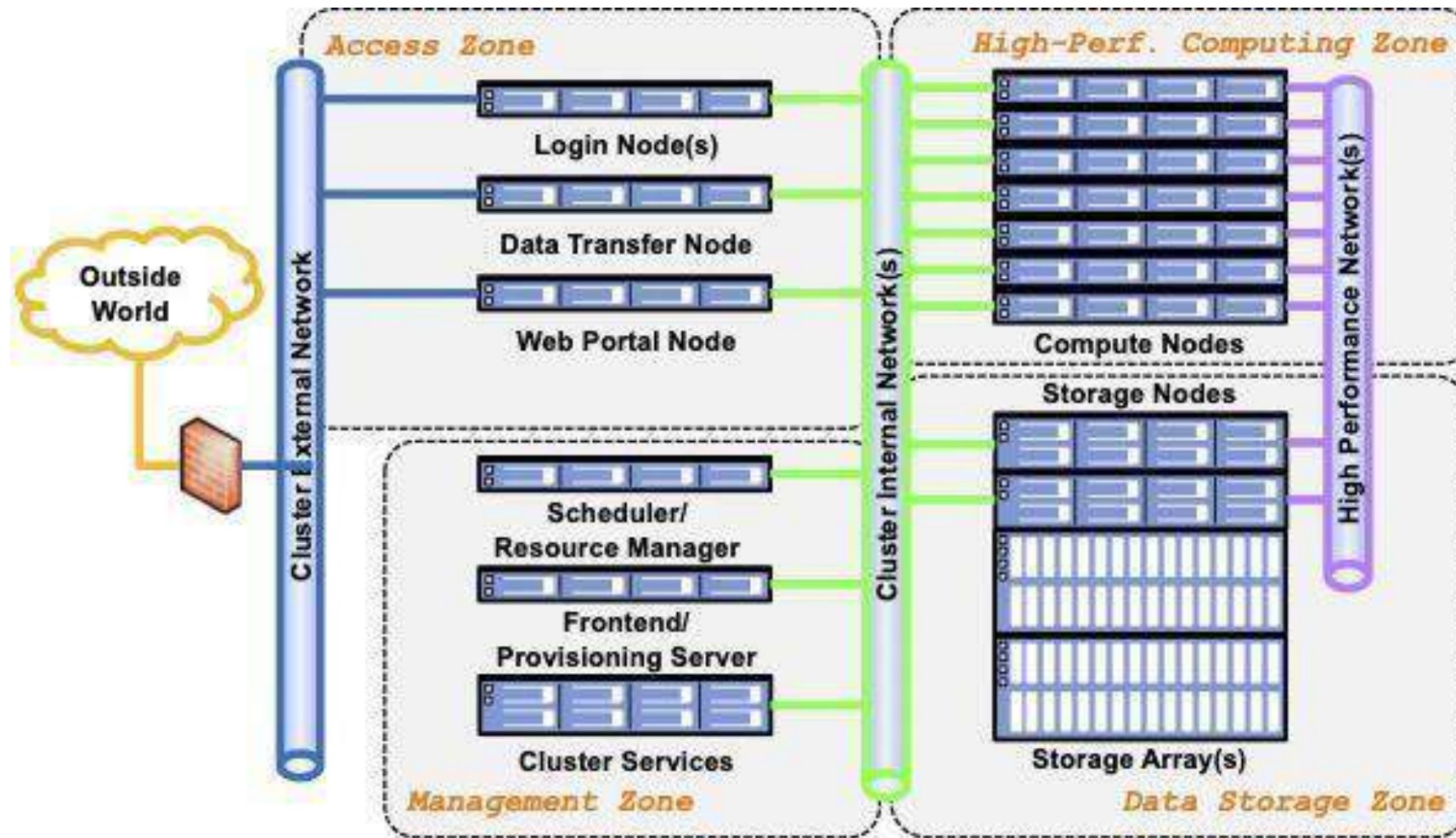
- Privilege is granted to processes based on need
- User processes live in the outermost rings with the least privilege
- System processes live closer to or at the center
- Interfaces between Rings are important to protect from malicious users



Attribution:

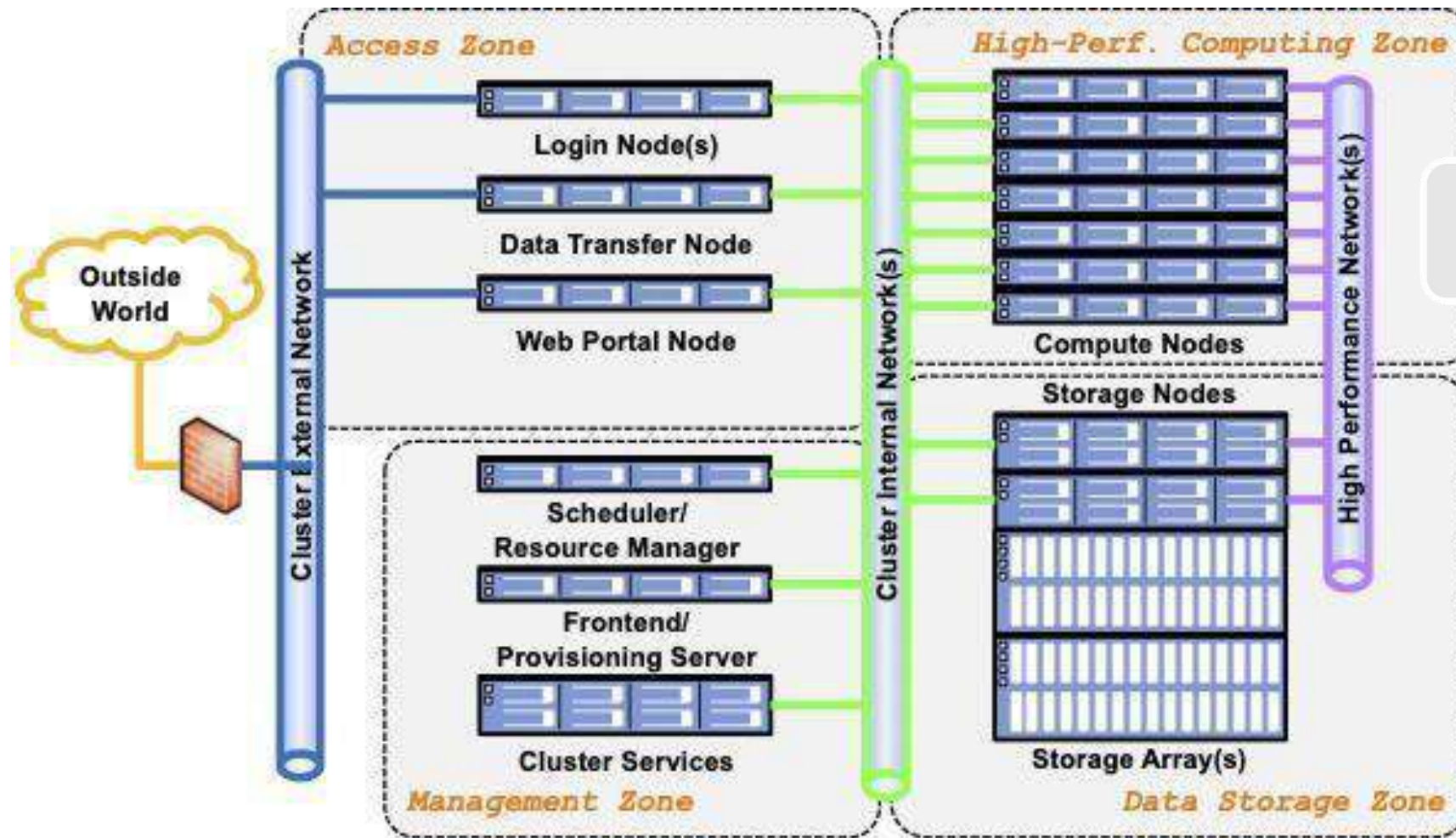
[Hertzprung](#) at [English Wikipedia](#)

NIST SP 800-223: HPC Security Architecture



Guo Y, Chandramouli R, Wofford L, Gregg R, Key G, Clark A, Hinton C, Prout A, Reuther A, Adamson R, Warren A, Bangalore P, Deumens E, Farkas C (2023) High-Performance Computing (HPC) Security: Architecture, Threat Analysis, and Security Posture. (National Institute of Standards and Technology, Gaithersburg, MD), NIST Special Publication (SP) NIST SP 800-223 ipd. <https://doi.org/10.6028/NIST.SP.800-223> ipd

NIST SP 800-223: HPC Security Architecture



Where is the **Reference Monitor** in this diagram?

Ramaswamy Chandramouli
Lowell Wofford
Rickey Gregg
Gary Key
Antwan Clark
Catherine Hinton
Andrew Prout
Albert Reuther
Ryan Adamson
Aron Warren
Purushotham Bangalore
Erik Deumens
Csilla Farkas

This publication is available free of charge from:
<https://doi.org/10.6028/NIST.SP.800-223>

NIST NATIONAL INSTITUTE OF
STANDARDS AND TECHNOLOGY
U.S. DEPARTMENT OF COMMERCE

Guo Y, Chandramouli R, Wofford L, Gregg R, Key G, Clark A, Hinton C, Prout A, Reuther A, Adamson R, Warren A, Bangalore P, Deumens E, Farkas C (2023) High-Performance Computing (HPC) Security: Architecture, Threat Analysis, and Security Posture. (National Institute of Standards and Technology, Gaithersburg, MD), NIST Special Publication (SP) NIST SP 800-223 ipd. <https://doi.org/10.6028/NIST.SP.800-223.ipd>

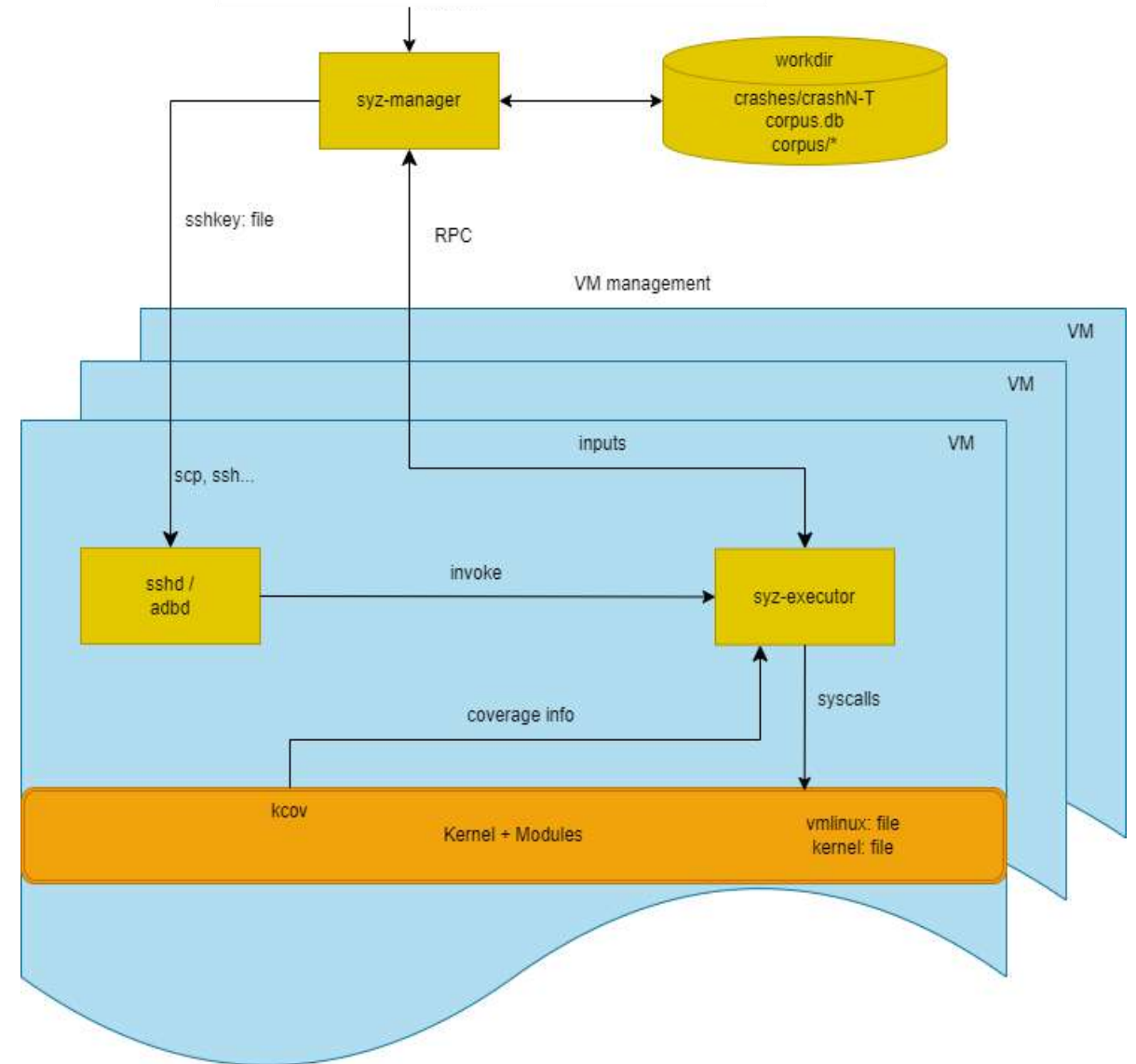
Syzkaller is a Coverage Guided Kernel Fuzzer

- **‘Kernel Fuzzer’**

- Quickly generates many small userland programs with widely varying inputs to the Kernel ABI layer through system calls
- Runs them against the Kernel ABI using system calls that are accessible to normal users

- **‘Coverage Guided’**

- Keeps traces of execution of the kernel, called a corpus
- Uses this information to more efficiently target certain areas



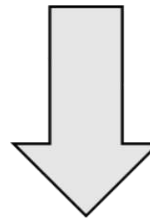
https://github.com/google/syzkaller/blob/master/docs/process_structure.png


```

r2 = openat$dirender128(0xffffffffffff9c, &(0x7f0000000000), 0x0, 0x0)
ioctl$DRM_IOCTL_SYNCOBJ_CREATE(r2, 0xc00864bf, &(0x7f0000000080)={<r3=>0x0})
ioctl$DRM_IOCTL_SYNCOBJ_TIMELINE_SIGNAL(r2, 0xc01864cd, &(0x7f0000000200)={&(0x7f0000000180)=[r3], 0x0, 0x1})
ioctl$DRM_IOCTL_SYNCOBJ_TRANSFER(r2, 0xc02064cc, &(0x7f0000000140)={r3, r3, 0x0, 0x81}) (rerun: 57)
ioctl$DRM_IOCTL_SYNCOBJ_TIMELINE_WAIT(r2, 0xc02864ca, &(0x7f0000000400)={&(0x7f0000000380)=[r3, r3], &(0x7f00000003c0)=[0x7000000000], 0x0, 0x2, 0x2})

```

SyzLang programs are converted to C code, compiled, and executed



```

#include <sys/syscall.h>
#include <sys/types.h>
#include <unistd.h>

uint64_t r[2] = {0xffffffffffffffff, 0x0};

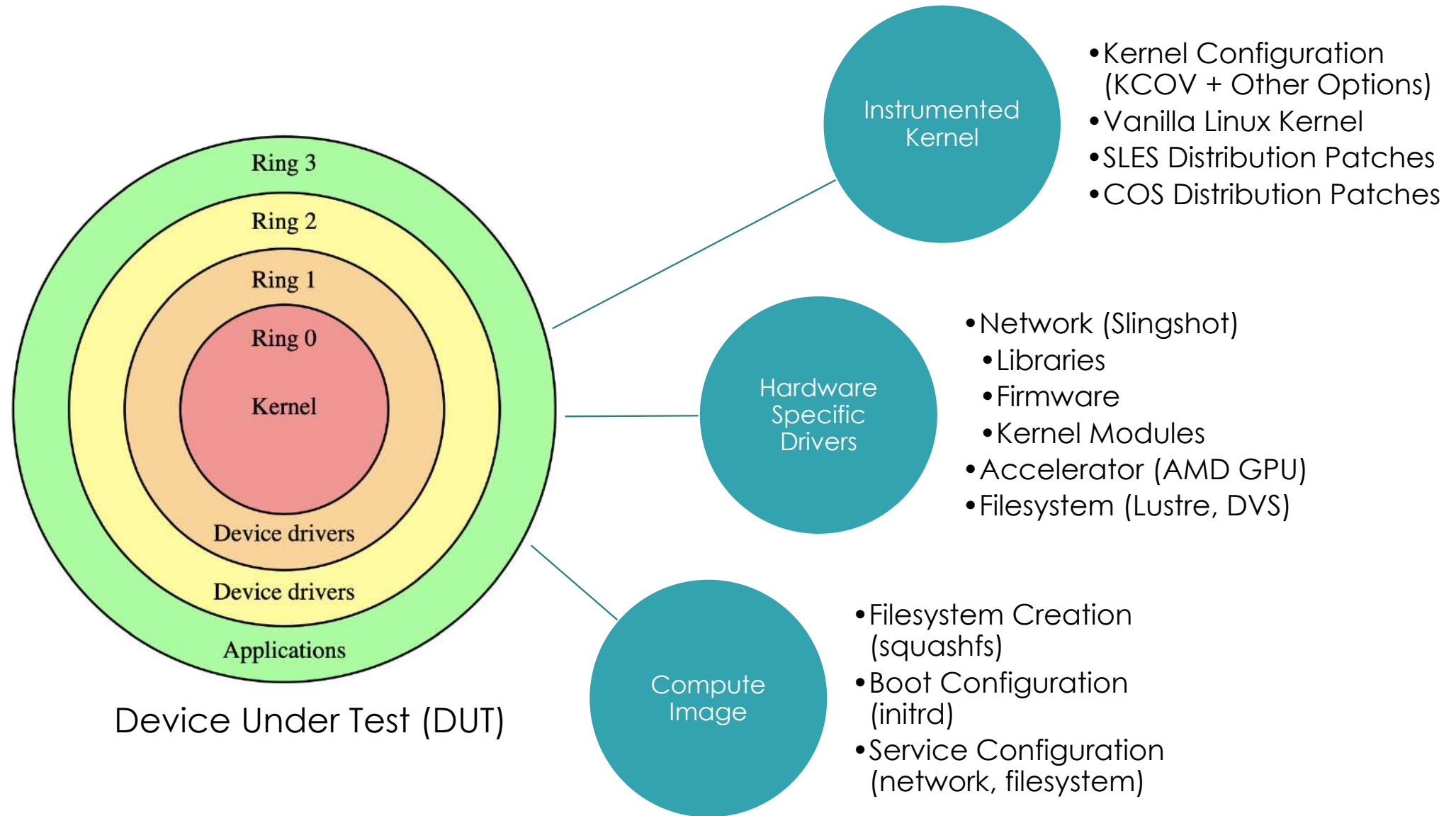
int main(void)
{
    syscall(__NR_mmap, 0x1ffff000ul, 0x1000ul, 0ul, 0x32ul, -1, 0ul);
    syscall(__NR_mmap, 0x20000000ul, 0x1000000ul, 7ul, 0x32ul, -1, 0ul);
    syscall(__NR_mmap, 0x21000000ul, 0x1000ul, 0ul, 0x32ul, -1, 0ul);
    intptr_t res = 0;
    memcpy((void*)0x20000000, "/dev/dri/renderD128\000", 20);
    res = syscall(__NR_openat, 0xffffffffffff9cul, 0x20000000ul, 0ul, 0ul);
    if (res != -1)
        r[0] = res;
    *(uint32_t*)0x20000084 = 0;
    res = syscall(__NR_ioctl, r[0], 0xc00864bf, 0x20000080ul);
    if (res != -1)

```

OS Trace of a Syzkaller Discovered Memory Flaw

```
[2023-07-25T17:07:40-04:00]-----[ cut here ]-----  
[2023-07-25T17:07:40-04:00]kernel BUG at mm/hugetlb.c:6941!  
[2023-07-25T17:07:40-04:00]invalid opcode: 0000 [#1] SMP NOPTI  
[2023-07-25T17:07:40-04:00]CPU: 20 PID: 40042 Comm: syz-executor Kdump: loaded Tainted: G          W OE  
[2023-07-25T17:07:40-04:00]Hardware name: HPE HPE_CRAY_EX235A/HPE CRAY EX235A, BIOS 1.6.2 03-22-2023  
[2023-07-25T17:07:40-04:00]RIP: 0010:huge_pte_offset+0x284/0x2b0  
[2023-07-25T17:07:40-04:00]Code: f7 c4 81 01 00 00 0f 84 c9 fe ff ff e8 d5 d8 e6 ff 41 81 e4 80 00 00 00 0f  
[2023-07-25T17:07:40-04:00]RSP: 0018:ffffc900b66b3c60 EFLAGS: 00010212  
[2023-07-25T17:07:40-04:00]RAX: 00000000000001c9 RBX: ffff88a0c6ce8000 RCX: fffffc9006930400  
[2023-07-25T17:07:40-04:00]RDX: 0000000000040000 RSI: ffffffff8132b344 RDI: ffff88a0c6ce8000  
[2023-07-25T17:07:40-04:00]RBP: 0000000020003000 R08: 0000000020000080 R09: 0000000000000002  
[2023-07-25T17:07:40-04:00]R10: fffffc900b66b3ef0 R11: 0000000000000000 R12: ffffffff82f842e0  
[2023-07-25T17:07:40-04:00]R13: 0000000000200000 R14: 0000000000000008 R15: 000000209d0c9067  
[2023-07-25T17:07:40-04:00]FS: 00007fffec800700(0000) GS:ffff88c04f700000(0000) knlGS:0000000000000000  
[2023-07-25T17:07:40-04:00]CS: 0010 DS: 0000 ES: 0000 CR0: 0000000080050033  
[2023-07-25T17:07:40-04:00]CR2: 000000000058bf8c CR3: 00000020c368a004 CR4: 0000000000770ee0  
[2023-07-25T17:07:40-04:00]DR0: 0000000000000029 DR1: 0000000000000000 DR2: 0000000000000000  
[2023-07-25T17:07:40-04:00]DR3: 0000000000000000 DR6: 00000000ffff0fff DR7: 0000000000000400  
[2023-07-25T17:07:40-04:00]PKRU: 55555554  
[2023-07-25T17:07:40-04:00]Call Trace:  
[2023-07-25T17:07:40-04:00] <TASK>  
[2023-07-25T17:07:40-04:00] follow_huge_pmd_pte+0xc6/0x1f0  
[2023-07-25T17:07:40-04:00] follow_page_mask+0x3fc/0x8d0  
[2023-07-25T17:07:40-04:00] ? futex_hash+0xe/0xd0  
[2023-07-25T17:07:40-04:00] follow_page+0x4d/0xe0
```

Building an Instrumented Cray OS HPC Node



Testing Configuration and Results

System Configuration

- Testing is performed on Borg, the Testing and Development System (TDS) for Frontier
- Kernel and driver source is used to build the Cray Operating System (COS) and related drivers
- Kernel Code Coverage (KCOV) is enabled in kernel build parameters during build time
- ~4 nodes are used during periods of fuzzing
- Filesystems are unmounted and network is usually isolated

Syzkaller Configuration

- Syzkaller is configured to operate in “Isolated” mode
 - 32 CPU Threads
 - ‘Setuid’ sandbox
- Compute node boot time: ~10-20 minutes
- Full compute image with replaced kernel and rebuilt initrd



Testing Configuration and Results

System Configuration

- Testing is performed on Borg, the Testing and Development System (TDS) for Frontier
- Kernel and driver source is used to build the Cray Operating System (COS) and related drivers
- Kernel Code Coverage (KCOV) is enabled in kernel build parameters during build time
- ~4 nodes are used during periods of fuzzing
- Filesystems are unmounted and network is usually isolated

Syzkaller Configuration

- Syzkaller is configured to operate in “Isolated” mode
 - 32 CPU Threads
 - ‘Setuid’ sandbox
- Compute node boot time: ~10-20 minutes
- Full compute image with replaced kernel and rebuilt initrd



On and off testing over 2 years led to **8 'consistent' crashes** and **4 reproducible** findings!
Vendor accepted one and assigned CVE-2025-27087

Discussion



Acknowledgements:

This research used resources of the Oak Ridge Leadership Computing Facility at the Oak Ridge National Laboratory, which is supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC05-00OR22725.